Evaluation Harness — ITDX26

Deterministic replay, signed run reports, and reproducibility discipline

Event	ITDX26 — U of Arizona Sierra Vista, 8–18 Sept 2026
Track	Algorithm-only, unclassified, laptop-based evaluation
Solicitation	W91RUSI-FCID260001 (SUB-007)
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Handling	UNCLASSIFIED // PROPRIETARY
Version	FormSpace overlay v1.0 — 2026-09-01

1. Four Tasks as Operations on Form Space

Task	Operation	Learned?	Live budget
12 Pattern	Class discovery in F_Λ	Yes (SSM/Mamba)	≤ 90 s
13 Link	Typed hypergraph on Form-classes	Yes (relational GNN)	≤ 120 s
8 COA	Trajectories on admissible submanifold	Yes (MYCA agents)	≤ 240 s
14 Map	Deterministic projection	No (deterministic)	≤ 30 s cold start

2. Ten Loss Terms (NLM training)

- Masked reconstruction.
- Forecast (multi-horizon).
- Contrastive alignment across modalities.
- Perturbation robustness.
- Physics regularization.
- Graph consistency.
- Calibration (Brier + ECE).
- Causal invariance.
- Safety (policy violation penalty).
- Composite λ-weighted total.

3. Reproducibility Discipline

- Five seeds: {11, 17, 23, 29, 31}.
- Grouped bootstrap by dataset-family (prevents leakage across NLM-Bench families).
- Paired significance tests on within-seed comparisons.
- Held-out data splits documented in run manifest.
- Every run produces a Merkle root; roots are chained across the 8-day event.

4. NLM Tier Sizes

Tier	Params	Use case	Laptop
Edge-small	10–80 M	Field devices, embedded	Both
Field-medium	80–350 M	Gateway, ITDX evaluation	Both
Foundation-large	350 M – 1.5 B	Stretch demo, offline	Apple 64 GB

Honest boundary: we do not pretrain a 2B–7B foundation model in the 8–24 h laptop window. We fine-tune existing tiers and report the exact base model, training compute, and data budget in the run manifest.

5. Signed Run Report — Contents

- Scenario YAML + SHA-256.
- Frozen NLM / FUSARIUM / MYCA container digests.
- Per-task metric CSV with 5-seed statistics.
- Per-task run log with timestamps.
- Merkle root of all observations consumed.
- SHA-256 list of all outputs.
- Fallback-state timeline (which mode was active when).
- Underperformance-statement log if any target was missed.

6. Cold-Machine Verification

Both laptops must pass a clean-machine test before Day 1: full disk decryption, USB mount, SHA-256 verification of the replay bundle, offline harness invocation, and a full replay run producing a Merkle root that matches the recorded root. This is the ITDX gate for entering the room.

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